RMIT Classification: Trusted

SSET Capstone Project
Student Self-Evaluation and Reflection

Please briefly respond to the following questions:

1. What are your main tasks and responsibilities within this Capstone project?

My main responsibility was developing and supporting the Bosch Rexroth virtual control system. This included learning and working with the ctrlX CORE Virtual environment, PLC programming, the Bosch 3D Viewer, and the web-based HMI. I also worked with the team on the system architecture and communication between the STM32, Simulink and the virtual Bosch system.

In addition, I contributed to the development and testing of the Stewart Platform model, including checking the inverse-kinematics outputs, testing different heave, roll and pitch commands, and helping with the integration and verification process. I also participated in PIL testing and the final testing, documentation and presentation preparation.

2. What are your completed works and contributions to the project until this point? From 0 to 10 (the higher the better), how would you rate your works and contributions?

I contributed to the development of the Bosch ctrlX CORE Virtual system, including the PLC control and 3D Viewer implementation. I also worked on the web-based HMI, which allows users to enter heave, roll and pitch commands and observe the Stewart Platform virtually.

I helped investigate Bosch documentation and understand how the ctrlX ecosystem, PLC variables, Data Layer and 3D Viewer could be used together. I also worked with Dung and Toan on the STM32–Simulink communication and participated in PIL testing. During the final stage, I helped test different input values in the virtual model, review the system limitations, prepare the final demonstration, and contribute to the report and presentation.

I would rate my contribution **8/10**. I completed my main assigned tasks and contributed across both the virtual Bosch system and the overall integration. However, the final communication between the physical STM32 system and the Bosch virtual system was not completed, so I believe there is still room for improvement in the integration work.

3. Briefly describe the main tasks and responsibilities of other student members in your project team. From 0 to 10 (the higher the better), how would you rate their works and contributions?

Pham Tri Dung: Worked mainly on the control algorithm, STM32 implementation and physical actuator control. He worked closely with Toan on implementing inverse kinematics on the STM32, stepper motor control, homing and actuator testing. **10/10.**

Le Huu Toan: Worked mainly on the embedded control and electrical side of the physical system. He contributed to STM32 programming, motor-driver control, limit switches, homing, actuator testing and hardware preparation. **10/10.**

Wong Vi Khang: Worked mainly on the Bosch virtual system, including PLC programming, the 3D Viewer and virtual platform control. He also contributed to testing and final demonstration preparation. **10/10.**

Pham Quan: Worked mainly on the mechanical and virtual-system development. He contributed to the platform design, industrial dimensions, PLC/3D Viewer development and testing. **10/10.**

Overall, all members contributed to different parts of the project, and the final result depended on combining the mechanical, embedded, PLC, HMI and simulation work.

4. Briefly describe critical issues, if any, that need to be addressed to further support the progression and completion of your project.

The main remaining issue is the lack of communication between the physical STM32 system and the Bosch ctrlX CORE Virtual system. The two systems were successfully demonstrated independently, but they were not connected into one real-time, bidirectional system.

Further development should establish a reliable communication protocol between the STM32 and ctrlX CORE, preferably through Ethernet/OPC UA, so that commands and physical feedback can be exchanged. The system should also use the Hall encoders in the linear actuators for closed-loop position feedback and perform more quantitative validation of leg accuracy, repeatability and digital-twin agreement.

The physical system should also be expanded to fully operate all three actuators together, with synchronized non-blocking motion and further safety validation. These improvements would move the current open-loop and partially integrated system closer to the original full HIL objective.